American International University-Bangladesh (AIUB)

Faculty of Science & Technology (FST)

Department of Computer Science

Introduction to Data Science

Mid-Term Project Report

Spring 2025-2026

Section: E

Group: 11

Project Contributions

SL #	Student Name & ID	Contributions (mention every part of the project where you contributed)
1.	Name:Rafi Ul Islam Rafi ID:22-49568-3	Coding ,preparing what things to do,graphs Handling missing values
2.	Name: Shuvo Chandra Mali ID: 21-45971-3	Coding,selecting topic,normalizing,Checking invalid values
3.	Name: Sadia Afrin Moumita ID: 22-49830-3	Loading the dataset,Removing duplicates
4.	Name: Israt Jahan Dristi ID: 22-49816-3	Creating subsets,Group comparisons,Train/test split

Dataset Description

It’s a dataset that shows data of 918 patients of different ages and gender who have heart failure issues,used to study and predict heart disease.

Theres a total of 12 coloumns of different sets of informations .they are :

Age - Patient’s age (integer).

Sex - Gender of the patient (factor with levels “M” and “F”).

ChestPainType -Type of chest pain (categorical: ATA, NAP, ASY, TA).

RestingBP -Resting blood pressure (numeric).

Cholesterol - Serum cholesterol level (numeric).

FastingBS - Fasting blood sugar (integer, usually 0 or 1).

RestingECG - Resting electrocardiogram results (categorical: Normal, ST, LVH).

MaxHR - Maximum heart rate achieved (integer).

ExerciseAngina - Exercise-induced angina (categorical: Y/N).

Oldpeak - ST depression induced by exercise relative to rest (numeric).

ST_Slope - Slope of the peak exercise ST segment (categorical: Up, Flat, Down).

HeartDisease - Target variable (factor with levels “0” = no disease, “1” = disease).

These features provide a mix of numerical and categorical data, making the dataset suitable for preprocessing, statistical analysis, and classification tasks.

Project Implementation Detail

Loading and Inspecting the Dataset

Description of the task1:

the dataset was loaded into R using the read.csv() function.

Code for task1

```
data <- read.csv("heart.csv")
head(data)
```

Sample output of your code for task1

```
> data <- read.csv("heart.csv")
> head(data)
  Age Sex ChestPainType RestingBP
1  40   M           ATA       140
2  49   F           NAP       160
3  37   M           ATA       130
4  48   F           ASY       138
5  54   M           NAP       150
6  39   M           NAP       120
  Cholesterol FastingBS RestingECG MaxHR
1         289         0     Normal   172
2         180         0     Normal   156
3         283         0         ST    98
4         214         0     Normal   108
5         195         0     Normal   122
6         339         0     Normal   170
  ExerciseAngina Oldpeak ST_Slope
1              N     0.0       Up
2              N     1.0     Flat
3              N     0.0       Up
4              Y     1.5     Flat
5              N     0.0       Up
6              N     0.0       Up
  HeartDisease
1            0
2            1
3            0
4            1
5            0
6            0
>
```

Output description of task1: The output loads the first six rows of the dataset proving the dataset is working.

Handling Missing Values

Description of the task2: check for missing values in the dataset and handle them.

Code for task2

colSums(is.na(data))

data$Cholesterol[is.na(data$Cholesterol)] <- mean(data$Cholesterol, na.rm=TRUE)

Sample output of your code for task2

```
> colSums(is.na(data))
            Age            Sex  ChestPainType      RestingBP    Cholesterol 
              0              0              0              0              0 
      FastingBS     RestingECG          MaxHR ExerciseAngina        Oldpeak 
              0              0              0              0              0 
       ST_Slope   HeartDisease 
              0              0 
>
> data$Cholesterol[is.na(data$Cholesterol)] <- mean(data$Cholesterol, na.rm=TRUE)
> |
```

Output description of task2: The output shows that all columns have zero missing values. For Cholesterol, if missing values were present, they would be replaced with the mean. This ensures the dataset is complete and ready for analysis.

Title of Task 3

Normalization of Continuous Variables

Description of Task 3: we normalize the continuous variable that are restingBP,Choleesterol,MaxHR, Oldpeak by min -max scaling process and bring all the values within range of 0 and 1 so that we can compare the values

Code for Task 3:

data$RestingBP <- (data$RestingBP - min(data$RestingBP)) / (max(data$RestingBP)-min(data$RestingBP))

data$Cholesterol <- (data$Cholesterol - min(data$Cholesterol)) /

(max(data$Cholesterol) - min(data$Cholesterol))

data$MaxHR <- (data$MaxHR - min(data$MaxHR)) /

(max(data$MaxHR) - min(data$MaxHR))

data$Oldpeak <- (data$Oldpeak - min(data$Oldpeak)) /

(max(data$Oldpeak) - min(data$Oldpeak))

Sample output of your code for task3

```
>
> data$Cholesterol[is.na(data$Cholesterol)] <- mean(data$Cholesterol, na.rm=TRUE)
> data$RestingBP <- (data$RestingBP - min(data$RestingBP)) / (max(data$RestingBP)-min(data$RestingBP))
>
> data$Cholesterol <- (data$Cholesterol - min(data$Cholesterol)) /
+   (max(data$Cholesterol) - min(data$Cholesterol))
>
>
> data$MaxHR <- (data$MaxHR - min(data$MaxHR)) /
+   (max(data$MaxHR) - min(data$MaxHR))
>
>
> data$Oldpeak <- (data$Oldpeak - min(data$Oldpeak)) /
+   (max(data$Oldpeak) - min(data$Oldpeak))
```

Output Description of Task 3 :

The code normalizes the coloumn values and sets them within 0and 1

Title of Task 4

Showing noramlized values

Description of Task 4

Here we are sowing the normalized values of the normalized coloumns

Code for Task 4

summary(data$RestingBP)

summary(data$Cholesterol)

summary(data$MaxHR)

summary(data$Oldpeak)

Sample output of your code for task4

```
> data$Oldpeak <- (data$Oldpeak - min(data$Oldpeak))
+    (max(data$Oldpeak) - min(data$Oldpeak))
> summary(data$RestingBP)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  0.000   0.600   0.650   0.662   0.700   1.000 
> summary(data$MaxHR)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
 0.0000  0.4225  0.5493  0.5409  0.6761  1.0000 
> summary(data$Oldpeak)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
 0.0000  0.2955  0.3636  0.3963  0.4659  1.0000 
> summary(data$Cholesterol)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
 0.0000  0.2873  0.3698  0.3297  0.4428  1.0000 
```

Output Description of Task 4

Here we see the min ,1st quitile,median,mean ,3rd quantile and max of all the normalized groups

Title of Task 5

Converting Categorical Variables

Description of Task 5

We convert categorical variables (Sex, HeartDisease) into factors so R treats them as categories instead of numbers.

Code for Task 5

data$Sex <- as.factor(data$Sex)

data$HeartDisease <- as.factor(data$HeartDisease)

Sample output of your code for task 5

```
> data$Sex <- as.factor(data$Sex)
> data$HeartDisease <- as.factor(data$HeartDisease)

$ Sex           : Factor w/ 2 levels "F","M": 2 1 2 1 2 2…
$ HeartDisease  : Factor w/ 2 levels "0","1": 1 2 1 2 1 1…
```

Output Description of Task 5

The output shows that Sex and HeartDisease are now factors with two levels.

Title of Task 6

Visualization

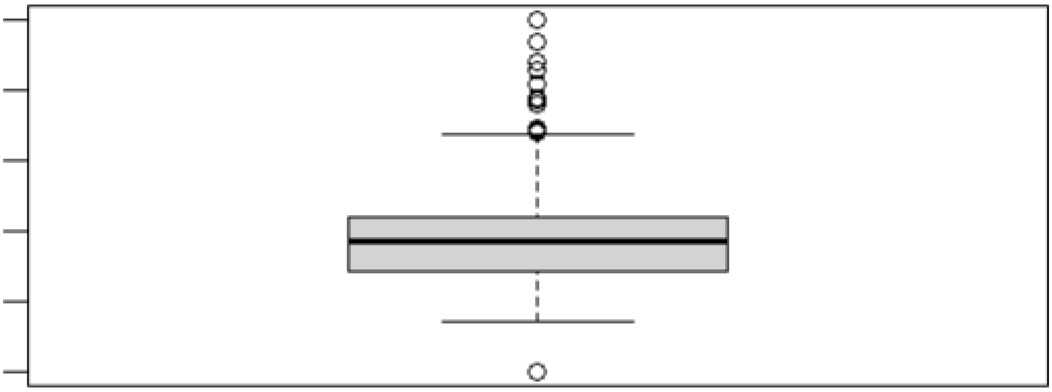
Description of Task 6

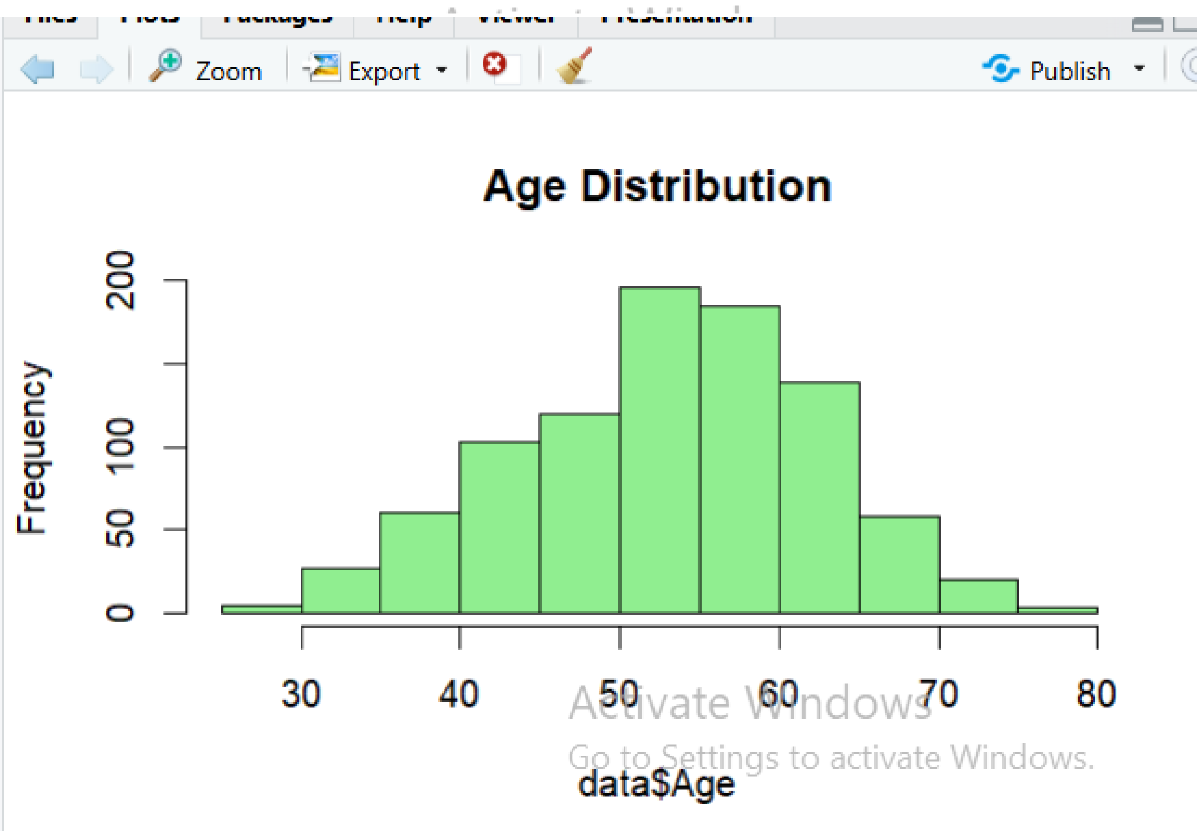
Making histograms and barplots to visualize distributions of Age and Chest Pain Types

Code for Task 6

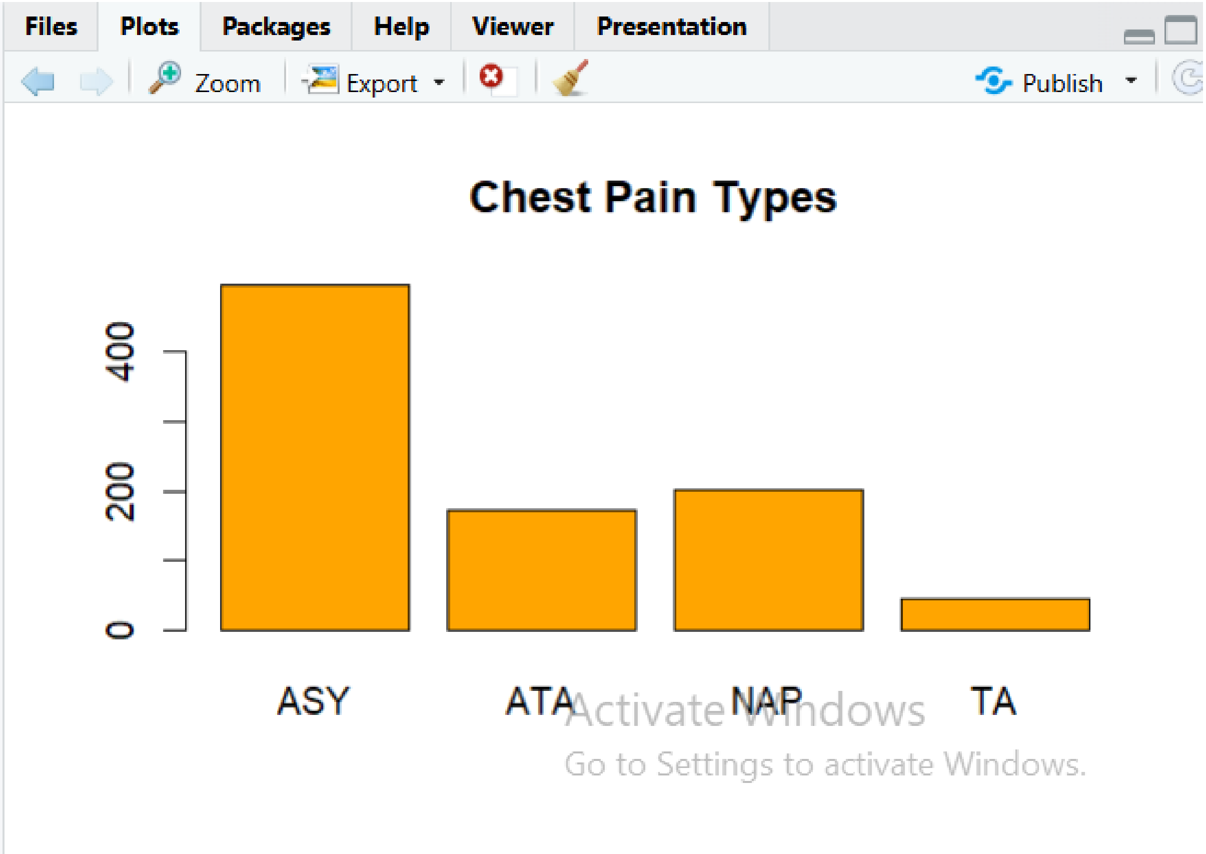
```
hist(data$Age, col="lightgreen", main="Age Distribution")

barplot(table(data$ChestPainType), col="orange", main="Chest Pain Types")

boxplot(data$Cholesterol)
```

Sample output of your code for task 6





Zoom
Export
Publish
Age Distribution
Frequency
0
50
100
200
30
40
50
60
70
80
data$Age
Activate Windows
Go to Settings to activate Windows.



Output Description of Task 6

The histogram reveals the age distribution of patients, while the barplot shows the frequency of different chest pain types. These visualizations help us understand the dataset better.

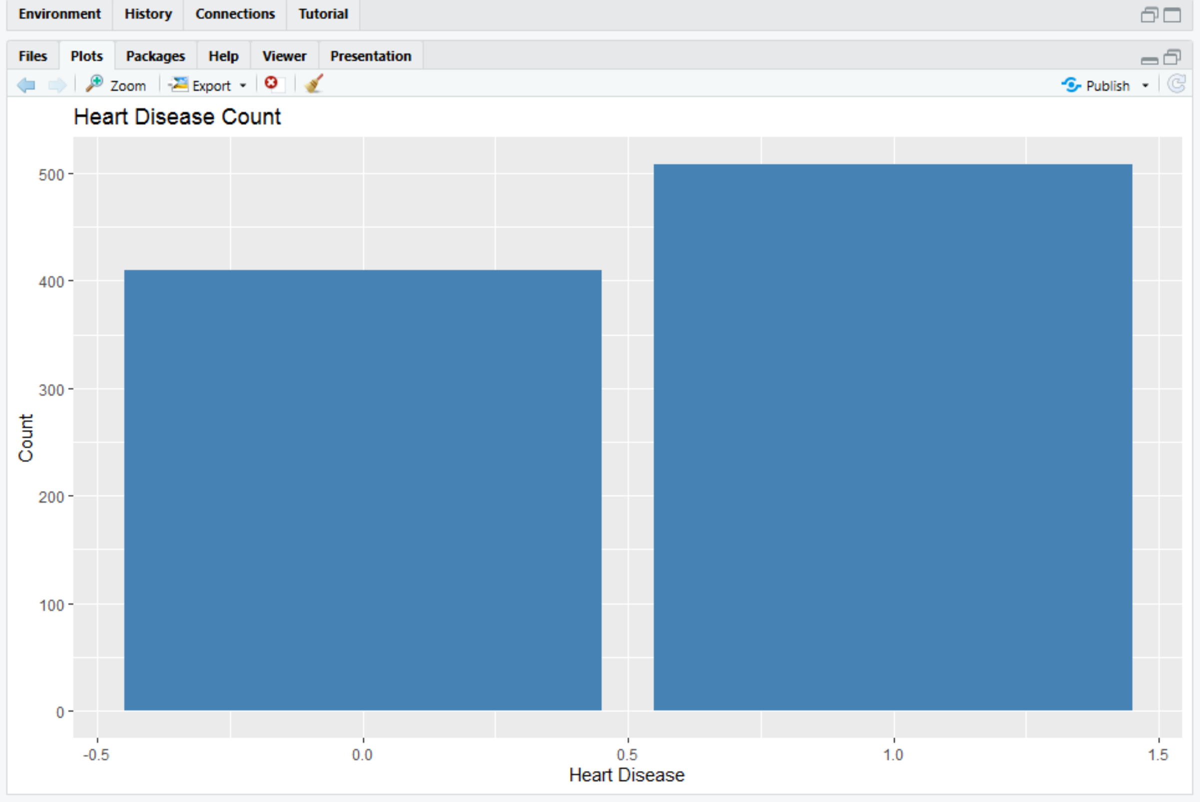
Task 7: Bar Plot of Heart Disease Count

Description of Task 7:
This graph shows how many patients have heart disease and how many do not.

Code:
library(ggplot2)

ggplot(data, aes(x = HeartDisease)) +

geom_bar(fill = "steelblue") + labs(title = "Heart Disease Count", x = "Heart Disease", y = "Count")



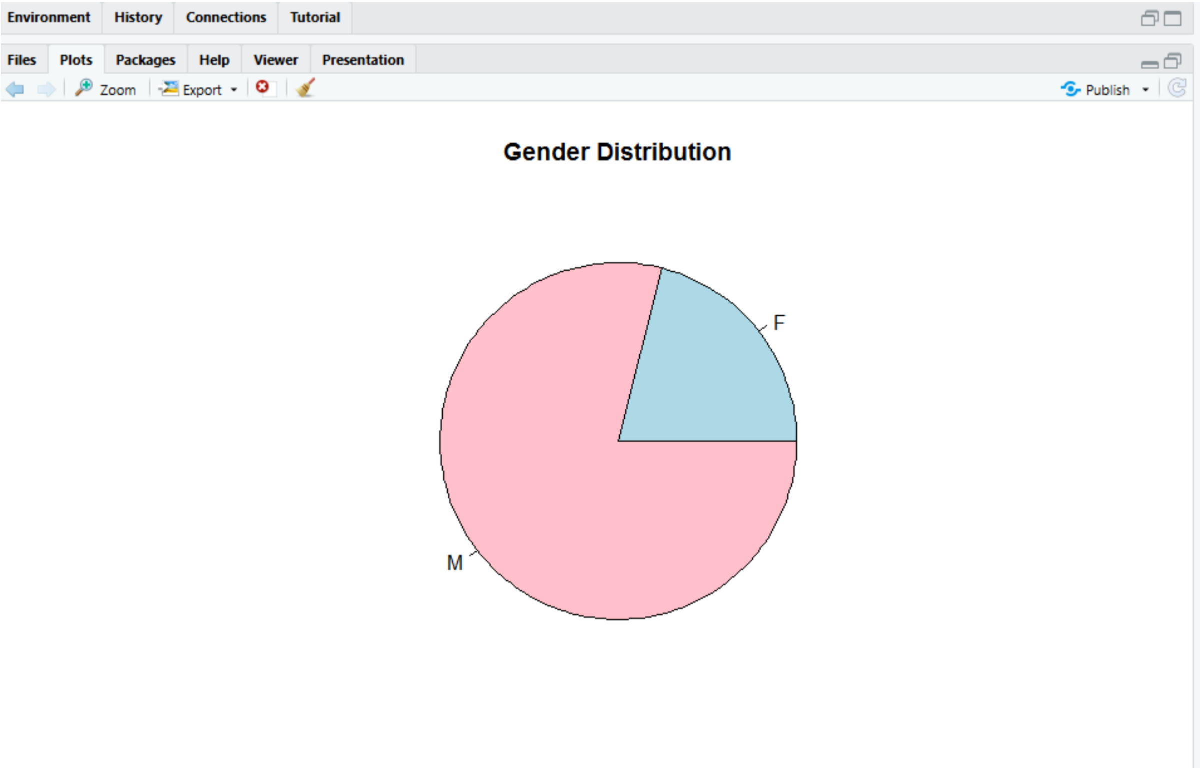
Code Description of Task 7:
geom_bar() counts the number of observations in each category (0 = No disease, 1 = Disease). This helps understand class distribution.

Task 8: Pie Chart of Gender Distribution

Description of Task 8:
This graph shows the proportion of male and female patients.

Code:

```
gender_count <- table(data$Sex)

pie(gender_count,

    main = "Gender Distribution",

    col = c("lightblue", "pink"))
```

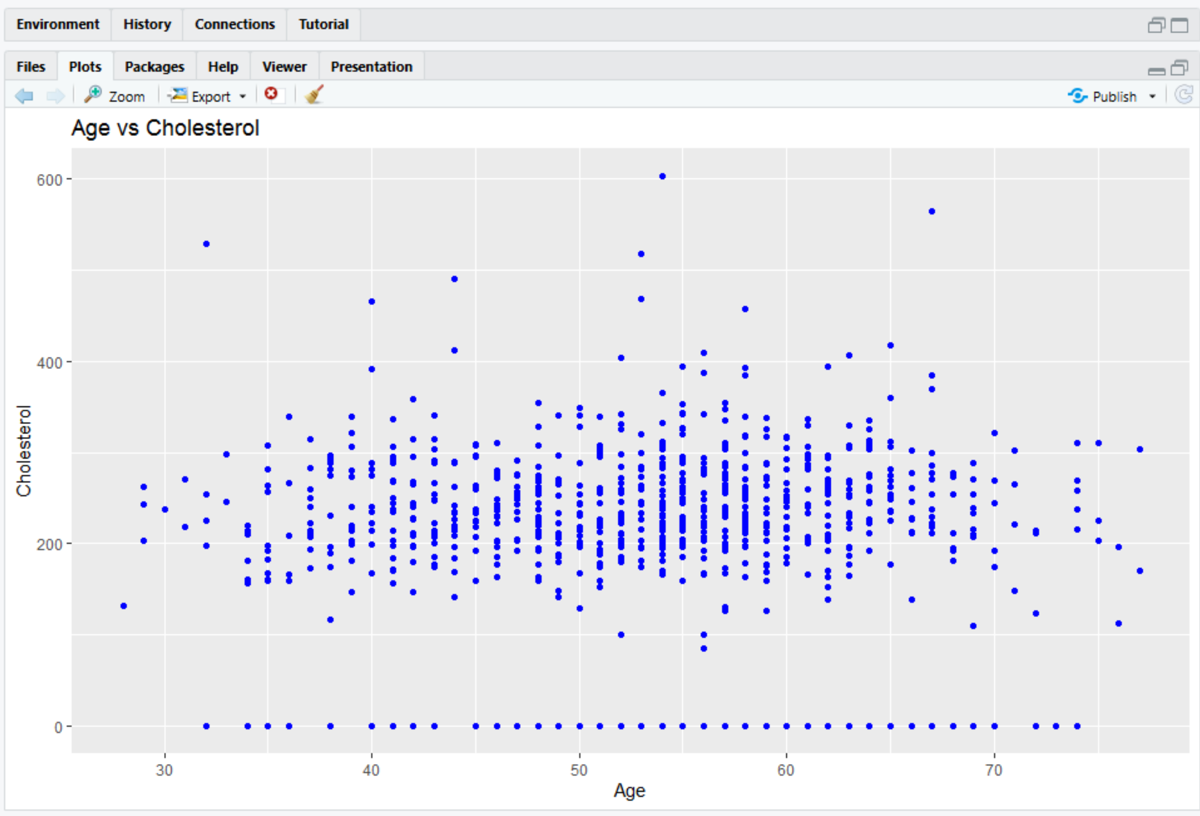


Code Description of Task 8:
table() counts frequency of each gender, and pie() visualizes their proportions.

Task 9: Scatter Plot (Age vs Cholesterol)

Description of Task 9:
This graph shows the relationship between age and cholesterol level.

Code:
```
ggplot(data, aes(x = Age, y = Cholesterol)) +

  geom_point(color = "blue") +

  labs(title = "Age vs Cholesterol")
```



Code Description of Task 9:
Each point represents a patient. This helps identify patterns or relationships between age and cholesterol.

Task 10: Calculating Mean, Median, and Standard Deviation

Description of Task 10:
This task calculates the mean, median, and standard deviation of important numerical variables such as Age, Cholesterol, and Max Heart Rate. These measures help to understand the central tendency and variability of the dataset.

Code:

```
mean(data$Age)

median(data$Age)

sd(data$Age)

mean(data$Cholesterol)

median(data$Cholesterol)

sd(data$Cholesterol)
```

mean(data$MaxHR)

median(data$MaxHR)

sd(data$MaxHR)

```
> mean(data$Age)
[1] 53.51089
> median(data$Age)
[1] 54
> sd(data$Age)
[1] 9.432617
>
> mean(data$Cholesterol)
[1] 198.7996
> median(data$Cholesterol)
[1] 223
> sd(data$Cholesterol)
[1] 109.3841
>
> mean(data$MaxHR)
[1] 136.8094
> median(data$MaxHR)
[1] 138
> sd(data$MaxHR)
[1] 25.46033
>
```

Code Description of Task 10:
The `mean()` function calculates the average value of a variable.
The `median()` function finds the middle value when data is sorted.
The `sd()` function computes the standard deviation, which shows how much the data varies from the mean.

- For **Age**, these values show the average age and how spread out the ages are.
- For **Cholesterol**, they indicate the typical cholesterol level and variation among patients.
- For **MaxHR**, they show the average maximum heart rate and its distribution.

These statistics help summarize the dataset and provide insights into the overall characteristics of patients.

Task 11: Correlation Between Variables

Description of Task 11:
Find the relationship between Age, Cholesterol, and MaxHR.

Code:

cor(data$Age, data$Cholesterol)

cor(data$Age, data$MaxHR)

cor(data$Cholesterol, data$MaxHR)

```
> cor(data$Age, data$Cholesterol)
[1] -0.09528177
> cor(data$Age, data$MaxHR)
[1] -0.3820447
> cor(data$Cholesterol, data$MaxHR)
[1] 0.2357924
> |
```

Code Description of Task:
`cor()` measures strength and direction of relationships between variables.